

Notes: Bellemare & Sebald (RES, 2023)

Measuring Belief-dependent Preferences without Data on Beliefs

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1 Big picture

- **Question.** How can researchers measure the causal importance of belief-dependent preferences, such as guilt aversion and reciprocity, when the researcher does not observe players' beliefs or higher-order beliefs?
- **Problem with standard practice.** A common empirical strategy elicits beliefs and then regresses choices on those beliefs. This can be misleading because stated beliefs can be rounded, strategically reported, or rationalized after choices.
- **Core insight.** In many psychological-game models, belief-dependent payoffs are unknown because beliefs are unobserved, but they lie inside intervals pinned down by the game payoffs. These payoff-implied intervals can be used to construct bounds on the sensitivity to belief-dependent preferences.
- **Experimental design.** The authors use two sets of sequential two-player games. Set I varies payoff consequences relevant for guilt aversion and reciprocity. Set II uses payoff-invariant games to estimate a nonparametric link between payoff differences and selfish choices when belief-dependent motives are neutralized.
- **Main parameter.** The key object is a game-specific sensitivity parameter ϕ_j that scales the psychological payoff from guilt aversion or reciprocity relative to monetary payoffs and idiosyncratic decision noise.
- **Main methodological result.** Without any belief data, the researcher can estimate an identification region for ϕ_j by combining observed choice shares in Set I with the estimated choice function from Set II.
- **Empirical result for guilt aversion.** Bounds for guilt aversion imply very large negative sensitivities. Point estimates using elicited beliefs fall inside the bounds but are implausibly large, suggesting that simple guilt aversion is not the best explanation for behaviour in this experiment.
- **Empirical result for reciprocity.** Bounds for reciprocity are tight, positive, and economically meaningful. They imply significant reciprocal preferences in the population.
- **Mixture of motives.** Post-experimental questionnaire data reveal that subjects self-report different motivations, especially selfishness and reciprocity. Subgroup analyses reinforce the evidence for reciprocity.
- **Validation.** The paper compares bounds to point estimates using elicited beliefs and finds that the point estimates lie inside estimated bounds. This suggests that elicited belief data in this experiment may not be severely distorted by reporting biases.
- **Takeaway.** The paper provides a partial-identification method for estimating belief-dependent preferences without belief data, turning a difficult measurement problem into a bounds problem using experimental payoff variation.

2 Incremental contribution of the paper

2.1 Relative to belief-elicitation approaches

- Earlier empirical work on guilt aversion and reciprocity typically elicits first- or higher-order beliefs and then tests whether those beliefs predict choices in the direction implied by psychological-game models.
- Bellemare and Sebald show that belief elicitation is not necessary for informative inference about the strength of belief-dependent preferences.
- This is important because stated beliefs can suffer from nonclassical measurement error, rounding, ex post rationalization, false-consensus-type reporting, or framing effects.
- Their method estimates *bounds* using only choice data, payoff variation, and structural restrictions from the preference model.

Incremental contribution: The paper moves the literature from *belief-based point identification* to *choice-based partial identification*. This is a methodological contribution because it allows researchers to study belief-dependent preferences in datasets where beliefs are missing or questionable.

2.2 Relative to psychological-game theory

- Psychological-game models often define utility as a function of actions, material payoffs, and beliefs about beliefs.
- These models are rich, but this richness makes empirical implementation difficult because higher-order beliefs are hard to observe credibly.
- The paper shows that prominent models share an invariance property: in carefully designed games, belief-dependent payoff components can be bounded or set to zero by design.
- This creates empirically useful restrictions without abandoning the theoretical structure.

Incremental contribution: The paper makes psychological-game models empirically usable under weaker data requirements. It shows how model primitives can generate inequality restrictions on preference parameters.

2.3 Relative to partial identification

- The paper adapts ideas from Manski and Tamer-style interval measurement to belief-dependent preferences.
- The unobserved belief-dependent payoff is treated as an interval-valued regressor.
- The link function mapping monetary payoff differences into choice probabilities is estimated from payoff-invariant games.
- Bounds are then constructed by inverting inequalities implied by the binary-choice model.

Incremental contribution: The paper introduces a new application of partial identification: bounding social-preference parameters when a theoretically essential regressor, higher-order beliefs, is unobserved.

2.4 Substantive contribution

- The empirical application shows significant reciprocity, consistent with participants' self-reported motivations.
- It finds very high implied guilt-aversion estimates, raising doubts about the simple guilt-aversion specification in this setting.
- It shows that comparing bounds with belief-based point estimates can be used as a diagnostic for elicited-belief quality.

Incremental contribution: The paper contributes evidence that reciprocity is a strong behavioural motive in the experimental games, while also showing how the bounding method can reveal when a belief-dependent model is empirically implausible.

3 Experimental design and data

3.1 Subjects and overall structure

- Subjects in Denmark participate in sequential two-player games.
- The analysis focuses on decisions made by B -players after A -players choose whether to let B decide the final allocation.

- Payoffs are denominated in Danish crowns (DKK).
- The design contains two main sets of games: Set I and Set II.
- Set I is used to learn about belief-dependent motives. Set II is used to estimate the payoff-based choice function when belief-dependent motives are neutralized by design.

Interpretation: The experiment separates the behavioural motive of interest from ordinary payoff sensitivity by including games in which belief-dependent terms vary and games in which they are shut down.

3.2 Set I: variant sequential games

- In Set I, player A first chooses between L and R .
- If A chooses R , the game ends with outside-option payoffs $(45, 30)$ for (A, B) .
- If A chooses L , player B chooses between l and r .
- Choosing l gives B a high payoff: B receives 210 and A receives 30.
- Choosing r gives B a lower payoff: B receives 150 and A receives z .
- The three Set I games vary only in z : $z = 60$ in Game 1, $z = 90$ in Game 2, and $z = 120$ in Game 3.

Interpretation: The selfish choice is l because it gives B 210 rather than 150. The non-selfish choice is r because it sacrifices 60 DKK for B and gives A a higher payoff. Increasing z makes the non-selfish option more beneficial for A and therefore more relevant for guilt aversion and reciprocity.

3.3 Set II: payoff-invariant games

- Set II contains 202 games.
- As in Set I, A can choose R and end the game with $(45, 30)$.
- If A chooses L , B chooses between l and r .
- In Set II, B's choice does not affect A's payoff: A receives the same x regardless of whether B chooses l or r .
- There are two values of x : 60 and 120.
- The payoff to B under l is 210, while the payoff under r is y , where y ranges from 150 to 250.

Interpretation: Because B's choice does not change A's payoff in Set II, the guilt-aversion and reciprocity terms studied in the paper are zero in these games. This makes Set II a calibration device for the monetary-payoff decision function.

3.4 Post-experimental motivation questionnaire

- After the experiment, subjects report their motivations.
- Categories include reciprocity, guilt aversion, inequity aversion, selfishness, and other motives.
- These stated motivations are not needed for the main bounds, but they are used for subgroup and finite-mixture-style analyses.

Interpretation: The questionnaire lets the authors check whether estimated patterns align with participants' self-reported motives and whether bounds remain informative in motivational subgroups.

4 Model framework

4.1 Utility specification

For a B-player in Set I game j , utility from action $a \in \{l, r\}$ is

$$u_B(a) = \pi_j^B(a) + \phi_j P(a, \pi_j, E_B(E_A(\pi_j^A, \pi_j^B))) + \epsilon_j(a). \quad (1)$$

- $\pi_j^B(a)$ is B's monetary payoff from action a .
- $P(\cdot)$ is the belief-dependent psychological payoff.
- ϕ_j is the sensitivity to the belief-dependent payoff.
- $E_B(E_A(\cdot))$ denotes B's belief about A's expectation.
- $\epsilon_j(a)$ is an unobserved action-specific payoff shock.

Interpretation: The parameter ϕ_j is the key object. It measures how strongly belief-dependent preferences affect decisions relative to money and unobserved shocks.

4.2 Choice probability

Let

$$\Delta\pi_j^B = \pi_j^B(l) - \pi_j^B(r) \quad (2)$$

and

$$\Delta P_j = P(l, \pi_j, E_B(E_A(\pi_j^A, \pi_j^B))) - P(r, \pi_j, E_B(E_A(\pi_j^A, \pi_j^B))). \quad (3)$$

If $s = 1$ denotes choosing the selfish option l , then

$$\Pr(s = 1 \mid \text{game} = j) = F(\Delta\pi_j^B + \phi_j \Delta P_j), \quad (4)$$

where F is the distribution function of the difference in unobserved payoff shocks.

Interpretation: This is a single-index binary choice model. The difficulty is that ΔP_j is unknown because beliefs are unobserved.

4.3 Interval measurement of the psychological payoff

Although ΔP_j is unobserved, the possible range of beliefs is known by design. Therefore

$$\Delta P_j \in [\underline{\Delta P_j}, \overline{\Delta P_j}]. \quad (5)$$

This implies

$$\Pr(s = 1 \mid \text{game} = j) \in [F(\Delta\pi_j^B + \phi_j \underline{\Delta P_j}), F(\Delta\pi_j^B + \phi_j \overline{\Delta P_j})]. \quad (6)$$

Interpretation: The unknown belief-dependent component becomes an interval-valued regressor. This is the source of partial identification.

4.4 Estimating the link function from Set II

In Set II, belief-dependent motives vanish:

$$\Delta P_k = 0. \quad (7)$$

Therefore the choice probability in Set II is

$$\Pr(s = 1 \mid \text{game} = k) = F(\Delta\pi_k^B). \quad (8)$$

- $F(\cdot)$ is estimated nonparametrically using choices in the 202 Set II games.
- The dense variation in $\Delta\pi_k^B$ identifies how monetary payoff differences map into selfish-choice probabilities.
- The estimated inverse link F^{-1} is used to build bounds for ϕ_j .

Interpretation: Set II transforms a difficult unobserved-belief problem into a standard nonparametric estimation problem.

4.5 Guilt aversion

In the simple guilt-aversion model, B dislikes letting down A. Choosing the kind action r creates no guilt payoff:

$$P(r, \cdot) = 0. \quad (9)$$

Choosing l creates guilt when A expected more than A actually receives:

$$P(l, \pi_j, E_B(E_A(\pi_j^A))) = E_B(E_A(\pi_j^A)) - \pi_j^A(l). \quad (10)$$

The possible psychological payoff difference satisfies

$$\Delta P_j \in [0, \pi_j^A(r) - \pi_j^A(l)]. \quad (11)$$

For guilt aversion, $\phi_j \leq 0$ because guilt reduces the attractiveness of the selfish action.

Interpretation: Increasing $z = \pi_j^A(r)$ raises the possible let-down associated with choosing l , so guilt aversion predicts less selfish behaviour from Game 1 to Game 3.

4.6 Reciprocity

In the Dufwenberg–Kirchsteiger-style reciprocity model, B’s psychological payoff is the product of perceived kindness by A and kindness by B:

$$P(a, \pi_j, E_B(E_A(\pi_j^B))) = PK_j \times K_j(a). \quad (12)$$

The equitable payoff for B is

$$\pi_j^e = \theta E_B(E_A(\pi_j^B)) + (1 - \theta)\pi_j^B(R). \quad (13)$$

A’s perceived kindness is

$$PK_j = E_B(E_A(\pi_j^B)) - \pi_j^e, \quad (14)$$

and B’s kindness from action a is

$$K_j(a) = \pi_j^A(a) - \pi_j^A(-a). \quad (15)$$

For reciprocity, $\phi_j \geq 0$: B receives utility from responding kindly to perceived kindness.

Interpretation: Because A gives B the opportunity to choose, B may interpret A’s decision as kind and reciprocate by choosing r . The larger A’s payoff gain from r , the stronger the reciprocal motive.

5 Identification and inference

5.1 Identification regions

Let

$$Q_j = F^{-1}(\Pr(s = 1 \mid \text{game} = j)). \quad (16)$$

The bounds come from the inequality

$$\Delta\pi_j^B + \phi_j \underline{\Delta P}_j \leq Q_j \leq \Delta\pi_j^B + \phi_j \overline{\Delta P}_j. \quad (17)$$

Solving this inequality gives a game-specific interval

$$\phi_j \in [\phi_{l,j}, \phi_{u,j}]. \quad (18)$$

Interpretation: The bounds contain all sensitivity parameters that can rationalize the observed choice share, given the model and all possible beliefs.

5.2 Bootstrap inference

- The authors use bootstrap confidence sets for the estimated identification regions.
- The bootstrap accounts for the two-stage procedure: estimating $F(\cdot)$ from Set II and then inverting Set I choice probabilities.
- They adapt the logic of Horowitz and Manski for inference on partially identified parameters.

Interpretation: Inference is about sets, not only point estimates. Confidence regions describe sampling uncertainty around the estimated interval.

5.3 Point estimates using elicited beliefs

The paper also elicits higher-order beliefs and computes point estimates for comparison. If μ_i^s is the elicited belief, it produces an individual-level ΔP_{ij} and

$$\Pr(s = 1 \mid \text{game} = j, i) = F(\Delta\pi_j^B + \phi_j \Delta P_{ij}). \quad (19)$$

Using the median relation,

$$\phi_j = \frac{Q_j^{Med} - \Delta\pi_j^B}{Med(\Delta P_{ij})}. \quad (20)$$

Interpretation: These belief-based point estimates are not the main estimator. They are used to compare with the choice-only bounds and diagnose whether elicited belief data behave plausibly.

5.4 Reporting bias diagnostic

- If elicited beliefs are strongly biased, point estimates may fall outside the choice-only bounds.
- Monte Carlo simulations show that rounding and reporting bias can push point estimates outside bounds.
- In the actual experiment, the point estimates fall inside the estimated bounds.

Interpretation: The bounding method can be used not only when belief data are absent, but also as a check on the credibility of belief data when beliefs are available.

6 Results and interpretation

6.1 Choice patterns and stated motives

- Selfish choices in Set I decline from Game 1 to Game 3.
- Self-reported selfish subjects choose the selfish option almost always.
- Self-reported reciprocal subjects choose the selfish option rarely.
- Guilt aversion is much less frequently reported than selfishness and reciprocity.
- Inequity aversion and other motives are present but less central to the main analysis.

Interpretation: The raw choice patterns already suggest that reciprocity and selfishness are major motives, while guilt aversion may not be the dominant explanation.

6.2 Guilt aversion bounds

- Estimated upper bounds for guilt aversion are negative in all three games.
- The all-subject upper bounds are approximately -2.144 in Game 1, -1.128 in Game 2, and -0.793 in Game 3.
- Belief-based point estimates are far more negative: around -8.359 , -3.857 , and -2.079 .
- Results are similar for men and women.

Interpretation: The results imply very strong guilt-aversion magnitudes. The authors view these magnitudes as implausibly high, suggesting that simple guilt aversion may be misspecified in this setting.

6.3 Reciprocity bounds

- Estimated reciprocity bounds are positive and tight.
- With $\theta = 0.5$, all-subject bounds are about $[0.012, 0.018]$ for Game 1, $[0.006, 0.009]$ for Game 2, and $[0.004, 0.007]$ for Game 3.
- Using the conservative approach with $\theta \in [0, 1]$ yields lower bounds but no finite upper bounds.
- For self-reported reciprocal subjects, the estimates are larger than for the full sample.

Interpretation: Reciprocity is both statistically and economically meaningful. The tight bounds make the reciprocity result the strongest substantive finding of the paper.

6.4 Validation with Experiment 2

- Experiment 2 provides additional data for invariant games.
- Figure 5 shows that functions estimated separately for self-declared selfish and reciprocal types are broadly consistent with the pooled function used in the main analysis.
- Replacing the original estimated $F(\cdot)$ with the function estimated from Experiment 2 gives similar results in supplementary analyses.

Interpretation: This supports the maintained assumption that the distribution of unobserved choice shocks can be transported from Set II to Set I.

7 Tables and figures

7.1 Figures 1 and 2: experimental games

Read:

- Figure 1 shows the three Set I games. The selfish option gives B 210 and A 30; the non-selfish option gives B 150 and A z .
- The only change across Set I games is $z \in \{60, 90, 120\}$.
- Figure 2 shows Set II games. B's choice changes B's payoff but not A's payoff.
- This makes Set II useful for estimating the choice response to monetary payoff differences when belief-dependent motives do not affect choices.

Economic message: The experimental design is the identification strategy: Set I creates belief-dependent payoff variation, while Set II identifies decision noise and monetary payoff sensitivity.

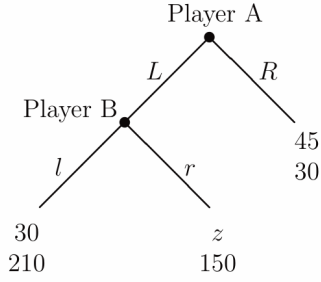


FIGURE 1

in games in *Set I*, where $z=60$ in “Game 1,” $z=90$ in “Game 2,” and $z=$

Figure 1: Set I games

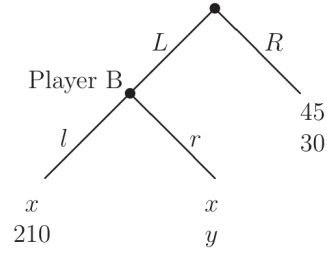


FIGURE 2

s in *Set II*. The first 101 games have $x=60$ and y takes 101 different values between 150 and 210. The last 101 games set $x=120$ and y takes 101 different values between 150 and 210.

Figure 2: Set II invariant games

7.2 Table 1: selfish choices and stated motives

Read:

- Across all B-players, selfish choices fall from 0.479 in Game 1 to 0.345 in Game 3.
- Self-reported selfish players choose selfishly almost always.
- Self-reported reciprocal players choose selfishly rarely.
- Guilt aversion is reported by only a small share of subjects.
- Men and women show similar qualitative patterns.

Economic message: The descriptive evidence suggests a finite mixture of motivations. Reciprocity and selfishness are the most prominent reported motives.

		By stated motives				
		Reciprocity	Guilt	Inequity	Selfish	Other
All B players						
Game 1	0.479 (467)	0.048 (145)	0.071 (14)	0.057 (69)	0.994 (177)	0.580 (62)
Game 2	0.413 (460)	0.030 (165)	0.091 (11)	0.029 (70)	0.965 (172)	0.381 (42)
Game 3	0.345 (463)	0.018 (167)	0.052 (19)	0.023 (85)	0.972 (144)	0.292 (48)
Total	0.412 (1390)	0.031 (477)	0.068 (44)	0.036 (224)	0.978 (493)	0.434 (152)
Share of motives	1.000	0.343	0.032	0.161	0.354	0.109
Men only						
Game 1	0.552 (134)	0.000 (39)	0.000 (3)	0.167 (12)	0.982 (57)	0.695 (23)
Game 2	0.443 (142)	0.008 (58)	0.000 (3)	0.000 (9)	0.948 (58)	0.285 (14)
Game 3	0.356 (160)	0.000 (57)	0.000 (4)	0.000 (25)	1.000 (52)	0.227 (22)
Total	0.444 (436)	0.026 (154)	0.000 (10)	0.043 (46)	0.976 (167)	0.424 (59)
Share of motives	1.000	0.353	0.023	0.106	0.383	0.135
Women only						
Game 1	0.450 (333)	0.066 (106)	0.091 (11)	0.035 (57)	1.000 (120)	0.512 (39)
Game 2	0.399 (318)	0.009 (107)	0.125 (8)	0.033 (61)	0.973 (114)	0.429 (28)
Game 3	0.339 (303)	0.027 (110)	0.067 (15)	0.033 (60)	0.956 (92)	0.346 (26)
Total	0.398 (954)	0.034 (323)	0.088 (34)	0.034 (178)	0.978 (326)	0.441 (93)
Share of motives	1.000	0.339	0.036	0.187	0.342	0.098

Notes: Numbers in parentheses represent sample sizes associated with each proportion. Proportions are presented (1) separately for each Game (rows denoted by “Game 1,” “Game 2,” and “Game 3”) as well as (2) pooled (row denoted by “Total”). Lower two blocks of the table present the same proportions split up by gender (blocks denoted by “Men only” and “Women only”). The table also reveals the share of B-players that indicated a certain motive in the post-experimental questionnaire (row denoted “Share of motives”).

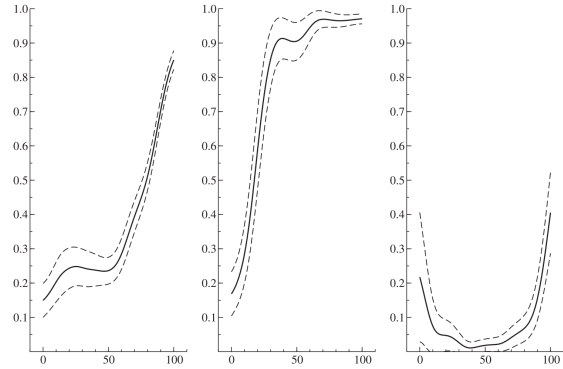


FIGURE 3

Left panel presents the nonparametric regression of the decision to choose the selfish option on second-order beliefs of all B players in Experiment 1. Middle panel shows corresponding estimates for self-declared selfish players, right panel shows corresponding estimates for reciprocal players. Estimated regression curves (full lines) and corresponding 95% confidence intervals (dashed lines) are presented. All estimates use the Gaussian kernel and bandwidth selected using Silverman’s rule.

7.3 Figure 3: elicited beliefs and selfish choices

Read:

- The left panel plots the nonparametric relation between choosing the selfish option and elicited second-order beliefs for all B-players.
- The middle panel shows a strong positive relation for self-declared selfish players.
- The right panel shows a less monotone pattern for self-declared reciprocal players.

Economic message: Belief-choice correlations can be suggestive but ambiguous. For selfish players, the positive relation is consistent with belief rationalization rather than genuine belief-dependent preferences.

7.4 Figure 4: invariant games and decision errors

Read:

- Figure 4 plots the probability of choosing selfishly in Set II as a function of B's payoff advantage from choosing selfishly.
- The two invariant-game subsamples, $x = 60$ and $x = 120$, show similar curves.
- This suggests that inequity aversion is not driving Set II choices.
- The pooled curve estimates the link function $F(\cdot)$ used in the bounds.

Economic message: Set II provides the empirical bridge from payoff differences to choice probabilities, allowing the authors to infer latent choice-noise quantiles without observing beliefs.

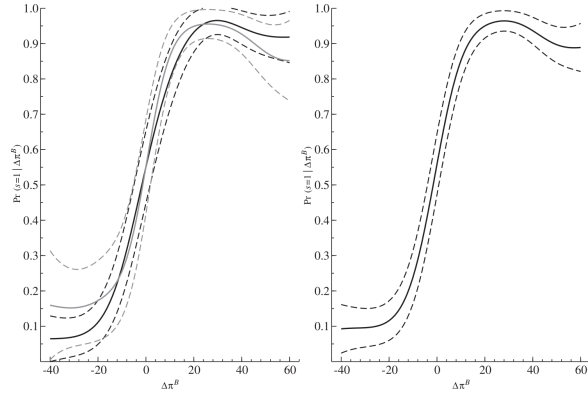


FIGURE 4

Non-parametric regression of s on $\Delta\pi^B$. Left panel presents the estimated regression curves (full lines) and corresponding 95% confidence intervals (dashed lines) for the subset of invariant games with player A payoff set to 60 (black) and the subset of games with player A payoff set to 120 (grey). Right panel presents the corresponding estimates obtained by pooling data from both subsets of invariant games. All estimates use the Gaussian kernel and bandwidths selected using Silverman's rule

TABLE 2
Interval and point estimates for guilt aversion

Guilt aversion	All		Men		Women	
	Interval	Point	Interval	Point	Interval	Point
Game 1	$(-\infty, -2.144]$	-8.359	$(-\infty, -2.026]$	-6.651	$(-\infty, -2.193]$	-8.482
	$(-\infty, -2.044]$	(0.745)	$(-\infty, -1.884]$	(1.081)	$(-\infty, -2.087]$	(0.833)
Game 2	$(-\infty, -1.128]$	-3.857	$(-\infty, -1.102]$	-4.346	$(-\infty, -1.140]$	-4.044
	$(-\infty, -1.072]$	(0.369)	$(-\infty, -1.052]$	(0.636)	$(-\infty, -1.084]$	(0.355)
Game 3	$(-\infty, -0.793]$	-2.079	$(-\infty, -0.785]$	-2.093	$(-\infty, -0.797]$	-2.004
	$(-\infty, -0.747]$	(0.159)	$(-\infty, -0.736]$	(0.248)	$(-\infty, -0.756]$	(0.207)

Notes: Bootstrap 95% confidence sets for the identification regions and bootstrap standard errors for direct point estimates in parenthesis. Estimates computed using proportions of selfish options (see Table 1 for sample sizes) and link function $F(\cdot)$ estimated using data from Set II games (right panel of Figure 4). Estimates reported using all subjects in each game, and separately for men and women in each game.

7.5 Table 2: guilt aversion estimates

Read:

- Bounds for guilt aversion are negative, as predicted by the model.
- The all-subject upper bounds are -2.144 , -1.128 , and -0.793 across Games 1–3.
- Point estimates using elicited beliefs are much more negative.
- Estimates are similar by gender.

Economic message: The negative bounds are consistent with guilt aversion, but the implied strength is implausibly large. This casts doubt on simple guilt aversion as the main behavioural model.

7.6 Table 3: reciprocity estimates

Read:

- With $\theta = 0.5$, reciprocity bounds are positive and precisely estimated.
- Conservative bounds with $\theta \in [0, 1]$ still have positive lower bounds but infinite upper bounds.
- Self-declared reciprocal subjects have larger estimates than the full sample.
- Point estimates using elicited beliefs lie inside the estimated bounds.

Economic message: Reciprocity is the clearest supported belief-dependent preference in the paper. The results are robust to gender splits and self-reported motivation splits.

Reciprocity	All		Men		Women		Reciprocal	
	Interval	Point	Interval	Point	Interval	Point	Interval	Point
$\theta = 0.5$								
Game 1	[0.012, 0.018] (0.011, 0.019)	0.012 (0.000)	[0.011, 0.017] (0.010, 0.018)	0.012 (0.001)	[0.012, 0.018] (0.011, 0.019)	0.013 (0.000)	[0.019, 0.028] (0.008, 0.038)	0.022 (0.000)
Game 2	[0.006, 0.009] (0.006, 0.010)	0.007 (0.000)	[0.006, 0.009] (0.006, 0.010)	0.007 (0.000)	[0.006, 0.010] (0.006, 0.011)	0.007 (0.000)	[0.009, 0.014] (0.004, 0.018)	0.011 (0.000)
Game 3	[0.004, 0.007] (0.004, 0.007)	0.005 (0.000)	[0.004, 0.007] (0.004, 0.007)	0.005 (0.000)	[0.004, 0.007] (0.004, 0.007)	0.005 (0.000)	[0.006, 0.009] (0.003, 0.012)	0.007 (0.000)
$\theta \in [0, 1]$								
Game 1	[0.006, + ∞] (0.005, + ∞)	0.012 (0.000)	[0.005, + ∞] (0.005, + ∞)	0.012 (0.000)	[0.006, + ∞] (0.006, + ∞)	0.013 (0.000)	[0.010, + ∞] (0.004, + ∞)	0.022 (0.000)
Game 2	[0.003, + ∞] (0.003, + ∞)	0.007 (0.000)	[0.003, + ∞] (0.003, + ∞)	0.007 (0.000)	[0.005, + ∞] (0.004, + ∞)	0.007 (0.000)	[0.005, + ∞] (0.002, + ∞)	0.011 (0.000)
Game 3	[0.002, + ∞] (0.002, + ∞)	0.005 (0.000)	[0.002, + ∞] (0.002, + ∞)	0.005 (0.000)	[0.002, + ∞] (0.002, + ∞)	0.005 (0.000)	[0.003, + ∞] (0.002, + ∞)	0.007 (0.000)

Notes: Bootstrap 95% confidence sets for the identification regions and bootstrap standard errors for direct point estimates in parenthesis. Estimates computed using proportions of selfish options (see Table 1 for sample sizes) and link function $F(\cdot)$ estimated using data from Set II games (right panel of Figure 4). Estimates reported using all subjects in each game.

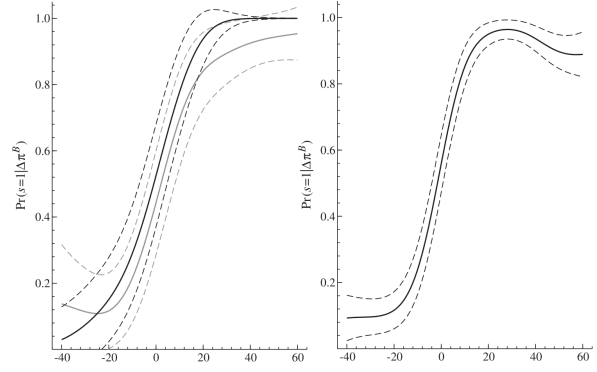


FIGURE 5

Left panel presents estimated nonparametric regression curves of s on $\Delta\pi^B$ in Experiment 2 (full lines) along with 95% confidence intervals (dashed lines) for self-declared selfish (black, $N=89$) and reciprocal (grey, $N=90$) B -players who played Set I games in Experiment 1. Right panel replicates the right-hand panel of Figure 4. All estimates use the Gaussian kernel and bandwidths selected using Silverman's rule.

7.7 Figure 5: validation using Experiment 2

Read:

- Figure 5 estimates the Set II choice function separately for self-declared selfish and reciprocal subjects in Experiment 2.
- The curves broadly agree over the relevant payoff range.
- The right panel replicates the pooled curve from Figure 4.

Economic message: Experiment 2 supports the assumption that the choice-error function is not strongly type-specific in the range needed for the bounds.

8 Short summary

- The paper studies how to measure belief-dependent preferences without observing beliefs.
- It focuses on guilt aversion and reciprocity in sequential two-player games.
- The design uses Set I games to vary psychological payoffs and Set II games to identify the payoff-choice link when psychological terms are zero.
- The method treats belief-dependent payoffs as interval-valued because beliefs are unobserved but payoff-implied belief ranges are known.

- The authors estimate bounds on sensitivity parameters rather than point estimates.
- The approach avoids reliance on elicited beliefs, which may be rounded or biased.
- Guilt-aversion bounds imply very large negative sensitivities and are viewed as implausible.
- Reciprocity bounds are tight, positive, and consistent with self-reported motivations.
- Comparing bounds with belief-based point estimates can diagnose belief-reporting problems.
- The main contribution is a choice-only partial-identification approach to belief-dependent preferences.

9 Conclusion

9.1 Core conclusion

Bellemare and Sebold show that researchers can measure belief-dependent preferences without collecting belief data. By exploiting payoff variation and the interval restrictions implied by psychological-game models, they estimate bounds on the causal effect of guilt aversion and reciprocity on choices.

9.2 Economic intuition

The key intuition is that the researcher does not need to know exactly what a player believes. The range of possible beliefs, combined with the payoff structure of the game, restricts the possible psychological payoff. These restrictions translate into bounds on the preference parameter. Payoff-invariant games then identify the choice-noise distribution needed to invert observed choice probabilities.

9.3 Incremental contribution

The paper’s incremental contribution is methodological and substantive. Methodologically, it provides a partial-identification approach for belief-dependent preferences that does not require belief data. Substantively, it shows that reciprocity is strongly supported in the experiment, while simple guilt aversion produces implausibly high estimates.

9.4 Takeaway

Belief-dependent preferences can be empirically studied even when beliefs are missing or potentially biased. Carefully designed choice data can deliver informative bounds, and those bounds can also be used to assess whether elicited beliefs are credible.